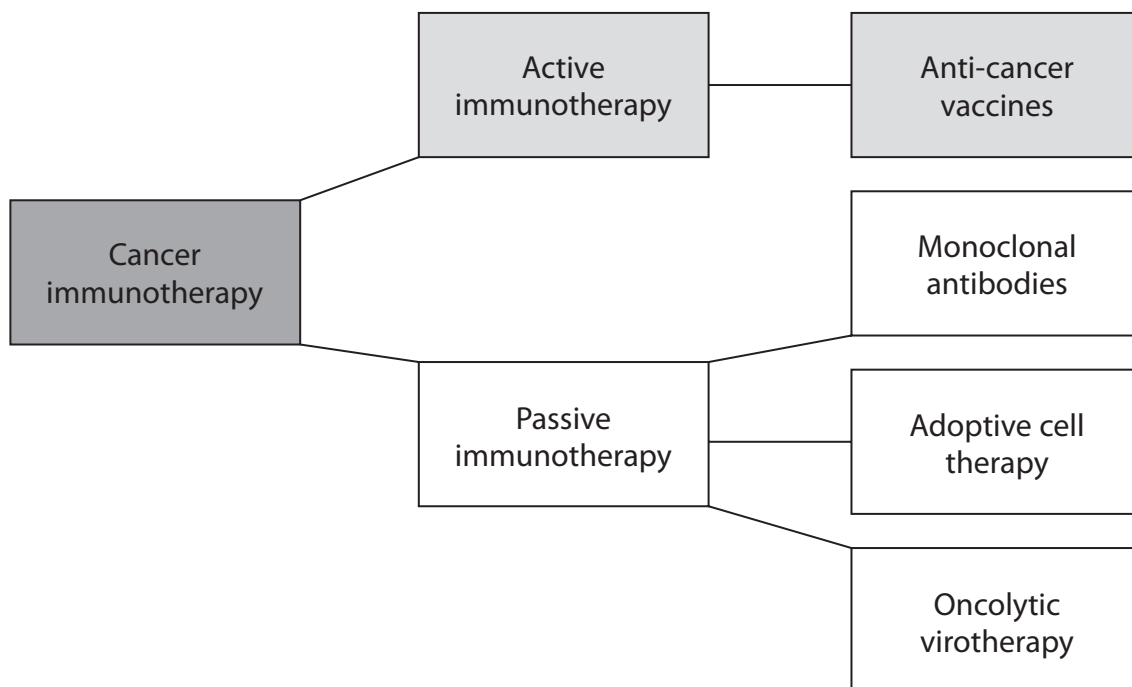


7 Cancer immunotherapy stimulates and strengthens an individual's active and passive immune systems to destroy cancer cells.

The diagram shows how different types of cancer immunotherapy can be classified.



(a) Suggest what the difference is between active immunotherapy and passive immunotherapy.

(2)

.....

.....

.....

.....

.....

.....

Some of these mutations can result in the formation of altered membrane proteins called neoantigens.

(i) Explain how a mutation can result in the formation of a neoantigen.

Explain how vaccines containing the mRNA coding for neoantigens stimulate the immune response.

This image shows a blank sheet of white paper with ten horizontal dashed lines spaced evenly apart, resembling a template for handwriting practice or a simple ledger. The lines are thin and grey, extending across the full width of the page. There is no text or other markings on the paper.

- (c) Monoclonal antibodies are made in a laboratory. Large numbers of identical antibodies specific to one type of antigen can be made.

Explain how these antibodies help the immune system to destroy cancer cells.

(2)

- (d) In adoptive cell therapy, T cells are taken from the patient and large numbers of genetically identical cells are made in the laboratory.

These cells are then injected back into the patient.

Explain why it is necessary to produce large numbers of genetically identical cells to make this treatment effective.

(2)

(e) In oncolytic virotherapy, viruses that destroy cancer cells are injected into the patient.

Not all types of virus are suitable for this therapy.

Suggest **two** features of a virus that make it suitable for use in this type of therapy.

(2)

(Total for Question 7 = 14 marks)
